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Autonomous Network Propagators: A Seismic Early-Warning Framework for AI-Driven Threat Propagation

Version 8.1 — Extended Publication Draft with Preliminary Synthetic Results

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Abstract

Autonomous propagation systems must learn their environment before they can damage it — and learning leaves traces. This paper reframes the Autonomous Network Propagator (ANP) framework around a predictive, two-phase model inspired by seismic early warning: an exploratory **P-phase**, in which the propagator expands its knowledge and reach through the environment graph and produces statistically detectable entropy expansion, followed by a structured **S-phase** of persistence, exfiltration, entrenchment, or disruption. Where earthquake early warning exploits the gap between fast precursor waves and slower destructive waves, ANP detection exploits the gap between in-band exploration and structural impact. The framework's central empirical claim is falsifiable and stated as a lead-time hypothesis: ANP detection must demonstrate measurable warning time before damage-phase propagation, beyond what existing lateral-movement detection provides.

Version 8.0 formalizes this model on measurable graph-theoretic quantities: an environment multigraph with declared-automation and ticketed-change attribution; a statistic vector comprising exploration-entropy gradient (breadth), weighted reach gain (value capture), and distributional divergence from declared baselines (shape); CUSUM accumulation against slow-and-low evasion; a human branching bound; and a predictive-coherence requirement linking exploration to impact. Fixed velocity thresholds from earlier versions are replaced by calibrated stochastic bounds, with the legacy thresholds retained as a fallback profile for graph-poor environments. Classification is split into **ANP-forecast** (precursor evidence, reversible response) and **ANP-confirmed** (impact plus predictive coherence), preserving early-warning value rather than reducing detection to post-hoc confirmation.

The six-clause behavioral definition, seven-primitive detection library, response governance with confidence bands and mandatory rollback, the network-observable ANP-N variant, governance and conformance mechanisms, and ethical boundaries of prior

versions are preserved and integrated — the seismic layer is the predictive core installed inside that structure, not a replacement for it. Two new primitives (entropy-gradient score; predictive-coherence score) extend the library, and the minimum-telemetry requirement drops to identity transitions, network adjacency, control-plane modifications, and change-ticket metadata. Throughout, the framework privileges interpretable, evidence-linked determinations over black-box scores, and states plainly that broad empirical validation remains an active research agenda whose core experiment — the lead-time hypothesis — this paper specifies.

Version 8.1 adds two elements reviewers of v8.0 identified as the path from specification to science: a formal exploration lower bound (Proposition 1) deriving, via Fano's inequality, the minimum in-band information acquisition any adaptive propagator must perform given its prior knowledge — converting the P-wave from plausible signature to bounded consequence of autonomy under uncertainty — and a preliminary synthetic study (n = 30 per scenario) demonstrating the pipeline end-to-end: median forecast lead of 4–5 windows before impact on epidemic and surgical scenarios, surgical detection requiring the full statistic vector (entropy-only: 0/30; full vector: 29/30), low false-forecast rates under quantile calibration (2–5 per 30 runs of 200 windows), and a measured rate-detectability frontier on slow-and-low variants that quantifies — rather than asserts — the patience cost imposed on evasive adversaries.

Keywords: autonomous malware, AI-powered threats, agentic AI security, early warning, entropy, lateral movement, zero trust, threat detection, interpretable detection

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1. Introduction

1.1 The Problem

The cybersecurity landscape is witnessing the emergence of a threat class that transcends traditional malware categories. As artificial intelligence and autonomous decision-making capabilities become broadly accessible — through hosted frontier models, open-weight models runnable on commodity hardware, and agent frameworks that wire models to tools — adversaries are positioned to deploy propagation systems that operate with reduced human intervention, adaptive target selection, and environmental reasoning. Such systems differ from historical worms and automated attack tools in three load-bearing ways:

1. **Consequential runtime decision-making.** The system makes choices — which target, which method, which path, when to act — based on observed environment state, rather than executing a fixed playbook.
2. **Multi-path propagation.** Movement is not confined to a single exploit vector; it flows opportunistically through credentials, APIs, orchestration layers, service trust relationships, and control-plane permissions.
3. **Behavioral coherence across distributed footholds.** New instances are not byte-identical copies, but they remain recognizable products of the same operating logic — the same shape of behavior, expressed across different contexts.

No inherited category captures this combination, and prior versions of this framework (v6, v7) answered the naming problem with a formal six-clause definition and a multi-plane detection library. Version 8.0 answers a harder criticism: *a taxonomy, however precise, detects after the fact.* Defenders do not primarily need a better name for what compromised them; they need warning before the compromise structures itself. This version supplies the predictive mechanism.

1.2 The Predictive Turn

The core observation is that autonomy has a thermodynamic signature. A system that selects its own targets and paths must acquire the information those selections require. Unless its knowledge of the environment was complete before it began — a condition addressed honestly in Sections 6.6 and 8.1 — it must explore: enumerate, probe, authenticate, traverse, and widen. Exploration is expansion of the propagator's empirical contact with the environment graph, and expansion of contact is measurable as entropy growth over observed transitions, branching beyond human baselines, and reach gain into weighted regions of the graph.

This motivates a two-phase model borrowed, as analogy rather than as physics, from seismology. Earthquake early-warning systems exploit the fact that fast-traveling P-waves arrive before destructive S-waves, buying seconds of actionable warning. Autonomous propagation exhibits an analogous structure for a different reason: not wave speed, but operational necessity — the exploratory phase (P-phase) is the propagator *learning*, and the damage phase (S-phase) is the propagator *spending* what it learned. The defender's opportunity is the gap between them. The analogy is pedagogical: in seismology both waves emanate from one rupture event; here the phases are sequential activities of one system. What transfers is the engineering pattern — detect the precursor, act before the destructive phase, and measure the system's value as *lead time*.

This reframing changes what the framework must prove. The class definition stands or falls with a falsifiable prediction: **entropy-based P-phase detection must yield measurable lead time before S-phase impact, and measurable blast-radius reduction when containment is keyed to it.** If exploration does not statistically predict impact — if the P-phase signal exists but does not anticipate where damage lands — the predictive layer fails its own test, and the framework reverts to the descriptive taxonomy of v7. That wager is stated as the core hypothesis of the validation program (Section 11).

1.3 Position of This Work

Three design commitments carry forward from prior versions, and a fourth is added:

- **Multi-plane corroboration.** No single telemetry plane — endpoint, identity, network, control-plane, or autonomy-trace — suffices for a strong determination. Strong claims require independent agreement across planes; this is simultaneously a false-positive control and an adversarial-robustness property.
- **Interpretability as a requirement, not a feature.** A determination must be expressible as: which clauses were satisfied, on which evidence, by which prim-

itives, at what confidence. Black-box anomaly scores may feed primitives; they may not constitute determinations.

- **Reversibility before aggression.** Every disruptive response primitive ships with a paired rollback path, and confidence bands gate aggressiveness. A framework that cannot recover from its own false positives will be abandoned after the first costly mistake.
- **Prediction over post-hoc categorization (new in v8).** Detection statistics are chosen for *lead time*, not only accuracy. A determination procedure that requires damage to have occurred before classifying the actor has forfeited the only advantage early detection offers; v8's forecast/confirm split (Section 5.8) exists to prevent exactly that failure.

1.4 Research Questions

RQ1. Can model-assisted reasoning over multi-plane telemetry reliably distinguish autonomous propagation from (a) conventional automation and (b) human-directed lateral movement?

RQ2. Which detection primitives — individually and in combination — contribute most to detection accuracy, and which multi-plane configurations achieve acceptable true-positive rates at the lowest false-positive cost?

RQ3. Does cross-plane reasoning (including LLM-based reasoning over normalized primitive outputs) outperform conventional SIEM-style correlation rules on time-to-detect and false-positive rate for ANP-class scenarios, including slow-and-low and mimicry variants?

RQ4. Do interpretable, evidence-linked explanations measurably reduce analyst investigation time and improve decision accuracy relative to opaque scoring, for surfaced ANP candidates?

RQ5 (new, core). Does P-phase entropy detection provide statistically significant lead time before S-phase structural impact, relative to lateral-movement detection baselines — and does containment keyed to ANP-forecast determinations measurably reduce S-phase blast radius?

RQ5 is the load-bearing question of v8: a negative result refutes the predictive layer and demotes the framework to taxonomy. Section 11 specifies the experiment.

1.5 Contributions

Version 8.0 makes six contributions, the first three new and the remainder integrated from prior versions:

1. **A seismic two-phase propagation model** — exploratory P-phase and structured S-phase — with a falsifiable lead-time hypothesis as the framework's central empirical claim.
2. **A graph-theoretic formalization** of P-phase detection: dual entropy quantities separating exploration from reach potential; a statistic vector (entropy gradient, weighted reach gain, distributional divergence) robust across epidemic, surgical, and mimicry expressions; CUSUM accumulation against slow-and-low evasion; a

formalized human branching bound; and a predictive-coherence condition with explicit null model.
3. **A forecast/confirm determination split** that preserves early-warning value: ANP-forecast triggers reversible response on precursor evidence; ANP-confirmed requires impact plus predictive coherence.
4. **The formal six-clause definition** with autonomy tiers, exclusions, and boundary cases — A4 (velocity) re-grounded on calibrated stochastic bounds with the legacy fixed thresholds retained as a fallback profile.
5. **A nine-primitive detection library** (seven carried, two new) with language-agnostic interfaces, declared failure modes, composition semantics, and an availability matrix — minimum telemetry reduced to identity transitions, network adjacency, control-plane modifications, and change-ticket metadata.
6. **An operational response framework** with phase-keyed actions, reproducible confidence bands, human-in-the-loop boundaries, mandatory rollback, the ANP-N variant, governance, conformance, and ethical boundaries.

1.6 Organization

Section 2 motivates the work against the present threat landscape. Section 3 reviews related work, states the categorical gap, and positions the early-warning analogies. Section 4 presents the six-clause definition. Section 5 — the core of v8 — formalizes the seismic propagation model and its detection statistics. Section 6 stress-tests the model adversarially. Section 7 specifies the detection primitive library. Section 8 covers response primitives with phase-keyed containment. Section 9 addresses telemetry integrity, governance, and ethics, and defines ANP-N and the profile system. Section 10 walks three vignettes through the full pipeline with phase annotations. Section 11 defines the validation program around the lead-time hypothesis. Section 12 states the research agenda; Section 13 concludes.

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2. Motivation: Why This Class, Why Now

2.1 The Capability Trajectory

Three independent trends converge on autonomous propagation as a near-term operational reality:

Model capability and availability. Frontier models demonstrate increasingly reliable multi-step tool use, environment exploration, code synthesis, and plan repair. Open-weight models lag the frontier but follow it, and run without provider-side safety enforcement or telemetry. The marginal cost of attaching a reasoning loop to an attack toolkit is collapsing toward the cost of inference.

Agent infrastructure. The orchestration patterns powering legitimate agentic software — tool registries, planner-executor loops, memory stores, recursive task decomposition — are general-purpose. An agent framework does not know whether the tool it invokes is a deployment script or an exploit. The engineering distance between an autonomous SRE assistant and an autonomous intruder is configuration, not architecture.

Defensive asymmetry at machine speed. Human-mediated detection and response operate on minutes-to-hours timescales. A propagator making even mediocre decisions at machine speed compresses the defender's reaction window below the human floor. It is not that autonomous propagators are smarter than human red teams — currently they are not — it is that they are *faster than human blue teams*, and they parallelize.

The predictive turn sharpens the third point. If the reaction window after impact begins is structurally too short for humans, the only window that matters is the one *before* impact — the P-phase. Early warning is not a convenience feature of the framework; it is the only regime in which human-governed response to machine-speed threats remains possible at all.

2.2 Early Indicators

Industry and government reporting has begun documenting model-assisted intrusion activity: AI-generated phishing and malware components, model-assisted vulnerability discovery, and agentic execution of multi-stage operations with human supervision at the objective level. We deliberately rest no claim on any single vendor report — attribution and capability claims in this space are contested — but the direction of travel is consistent across independent sources, and the constituent capabilities (autonomous target selection, credential-driven movement, control-plane manipulation, adaptive replanning) have each been demonstrated separately. ANP names their conjunction; v8 predicts their sequence.

A framework introduced after a threat class is widespread arrives too late to shape telemetry collection, detection engineering, or procurement. The window in which a definition can do the most good is precisely the window in which empirical incidence is still low — which is also the window in which it cannot be fully validated. Sections 9 and 11 treat this tension honestly.

2.3 The Cost of the Categorical Gap

The absence of a shared ANP-like category produces measurable defensive failures today:

- **Telemetry is not collected for autonomy signals.** Few environments preserve decision-loop structure, exploration topology, or change-ticket linkage at the granularity P-phase detection requires, because no detection category demands it. When the first major autonomous-propagation incident is investigated, the evidence most specific to the class will likely not exist.
- **Detections are plane-siloed.** Identity analytics, EDR, network analytics, and cloud-posture tools each see a slice of propagation behavior and alert independently. The defining behavior — coherent spread *across* planes, exploration *preceding* impact — is exactly what siloed, phase-blind tools cannot represent.
- **Benign automation pollutes the baseline.** Orchestration, autoscaling, patching, and CI/CD produce high-volume, machine-driven, multi-context activity superficially resembling propagation. Without declared-automation attribution built into the definition, defenders choose between alert floods and blindness.

- **Incident language fragments.** Post-incident reports describe the same behavior as "worm-like activity," "automated lateral movement," "AI-assisted intrusion," or "APT behavior," obstructing cross-incident learning.

The framework is engineered against all four failures: it names the telemetry that matters (and v8 reduces the minimum set), defines cross-plane corroboration as the unit of detection, builds benign-automation exclusion into both the definition and the statistics, and provides a versioned vocabulary for reporting — now including phase vocabulary.

---

3. Related Work and the Categorical Gap

3.1 Malware Propagation Models

Historical worm research modeled propagation as a stochastic epidemic process characterized by scanning behavior, infection curves, and signature-detectable payloads [4, 5, 13]. These models remain valuable at scale, but their assumptions — uniform exploitation, indiscriminate targeting, velocity as the dominant signal — are precisely what an adaptive propagator violates. A system that selects three high-value targets over six hours is invisible to an epidemic model tuned for three hundred targets in six minutes.

3.2 Lateral Movement Detection

Lateral-movement detection spans traffic analysis, authentication-pattern monitoring, and behavioral analytics [6, 7], with MITRE ATT&CK cataloguing the technique space [8]. The implicit actor model is a human operator or a fixed tool: detections key on known tool artifacts, known credential-abuse patterns, and human-timescale rhythms. Autonomous decision loops break these assumptions. ANP does not replace ATT&CK — ATT&CK remains the vocabulary of individual actions; ANP supplies the behavioral definition of the *actor class* executing them, a determination procedure for recognizing it, and (in v8) a predictive phase structure existing technique taxonomies do not express.

3.3 AI-Powered Threats and Agentic Security

A growing literature documents AI-powered offensive capability [9, 10], while parallel work on securing AI systems — prompt injection, tool poisoning, agent hijacking — is catalogued in MITRE ATLAS [14] and the OWASP LLM Top 10 [15]. The first literature treats AI as a tool augmenting existing attack categories; the second treats AI as a vulnerable asset. ANP addresses the configuration both imply but neither defines: AI as the operating logic of the threat itself, with the network as its medium.

3.4 Kill Chains and Campaign Models

Kill-chain models [16] describe staged human-directed operations. ANP's clause structure is deliberately stage-free with one principled exception: the P/S phase dis-

tinction (Section 5) is not a campaign stage model but a *learning/spending* dichotomy grounded in the information requirements of autonomy. An adaptive system may interleave canonical kill-chain stages arbitrarily; it cannot spend environmental knowledge it has not acquired.

3.5 Early Warning in Other Domains

The engineering pattern v8 adopts has precedents outside security. **Earthquake early warning** exploits the P/S wave-speed gap to issue alerts in the seconds between precursor detection and destructive arrival [17]. **Statistical process control** uses CUSUM charts [18] to detect small persistent shifts that per-sample thresholds miss — the formal tool v8 deploys against slow-and-low evasion. **Epidemic surveillance** uses syndromic precursors to trigger response ahead of confirmed outbreak. The common structure: a destructive process with an obligatory precursor, a detector tuned to the precursor rather than the destruction, and a value measure denominated in lead time. v8 instantiates this structure for autonomous propagation, with Section 5 supplying the precursor's formal signature and Section 11 the lead-time experiment.

3.6 Zero Trust and Defense-in-Depth

Zero trust architecture [2] and defense-in-depth [1] provide the structural substrate this framework assumes: segmented environments, continuous verification, layered telemetry. They constrain where a propagator can move and what evidence movement leaves, but do not define what autonomous propagation is, how to score it, or when to act. ANP is designed to be installed inside these architectures — and the v8 environment graph (Section 5.1) is, in zero-trust terms, simply the live map of the trust relationships those architectures are supposed to minimize.

3.7 Privacy-Preserving Collaborative Detection

Federated learning and secure aggregation enable cross-organizational detection without raw-data exchange [11, 12], directly informing the cross-tenant correlation primitive and the ANP-N exchange format: systemic awareness of a propagating class is a collective-action problem, and summarized signals must suffice where raw traces would otherwise be demanded.

3.8 The Categorical Gap, Stated Precisely

Why is ANP a distinct category rather than a repackaging of worms, lateral movement, botnets, credential abuse, and APT behavior? Because no existing category requires the conjunction, and the conjunction is what matters operationally:

Existing category	What it captures	What it discards
Worm	Self-propagation, velocity	Autonomy of target/path *selection*; selectivity; coherence under polymorphism
Lateral movement	Boundary-crossing technique	Actor model: assumes human direction or fixed tooling
Botnet	Distributed footholds, coherence	Assumes C2 dependency; ANP requires none
Credential abuse	Identity-plane technique	Single-plane; no propagation or autonomy semantics
APT	Adaptive, goal-directed campaigns	Human tasking at step level; human-timescale rhythms
Benign automation	Machine-driven multi-context activity	Declared, authorized, predictable; no exploit intent

The definition (Section 4) is a conjunction of six clauses such that deleting any one collapses the category into a row above. v8 adds a second, stronger distinctness argument: none of the inherited categories *predicts*. Each describes a behavior already underway; none asserts an obligatory precursor whose detection buys lead time. If RQ5 validates, ANP is distinct not merely as a definition but as a forecasting model — and if RQ5 fails, that result is publishable evidence that the conjunction lacks predictive value, which is the falsifiability the framework promises.

Box 1 — Why this is not just better lateral-movement detection:

Question	Lateral-movement detection	ANP v8
Detects exploration before impact?	No — alerts on movement techniques in progress	Yes — P-phase forecast is the design objective (hypothesis under test, RQ5)
Actor model includes runtime autonomy?	Human operator or fixed tooling assumed	A1 with tiers; behavioral autonomy evidence (branching bound) where traces absent
Requires predictive coherence?	No — alerts are independent events	Yes — impact must overlap exploration beyond a hypergeometric null (§5.6)

Question	Lateral-movement detection	ANP v8
Separates potential from acquired reach?	Rarely; often conflated in risk scores	Explicit: W_R/s_pp (potential, response-only) vs ΔW_R (acquired, evidence)
Falsifiable lead-time claim?	None stated	H-LEAD with stated refutation conditions (§11.2)
Cross-plane corroboration required for action?	Tool-dependent, usually optional	Mandatory for medium/high bands (§8.2)
Benign automation handled by	Allowlists, tuning	Declared-inventory attribution with residual statistics (§5.1)

---

4. Definition of Autonomous Network Propagators

4.1 Preliminaries and Notation

Let an **observation window** $W = [t_0, t_1]$ be a bounded interval of normalized telemetry. Let a **context** $c \in V$ be a unit of execution or control — host, container, account, service, cluster, tenant — and a **foothold** be durable execution or control capability within a context. Let S denote the software system under evaluation, observed as a correlated event stream E(S, W) spanning one or more telemetry planes P = {endpoint, identity, network, control-plane, autonomy-trace}. The environment graph G(t) and its derived quantities are defined in Section 5.1.

S is classified relative to predicates A1–A6. Clauses are individually necessary; jointly sufficient. Implementations must evaluate clauses independently and report per-clause evidence — a determination is the *explained conjunction*, never an unstructured score. v8 adds phase-resolved determination states (forecast/confirmed; Section 5.8) on top of the clause logic.

4.2 The Six Clauses

A1. Autonomy. S selects at least one of its targets, exploitation methods, or propagation paths through automated decision logic, without direct human selection at each step — via model-assisted reasoning, search, planning components, adaptive policies, or other automation. Two tiers:

- **A1-minimum:** S executes without per-step human selection and performs at least one consequential runtime choice conditioned on observed environment state.
- **A1-adaptive:** S updates or varies its target, method, timing, or path selection in response to observed environment state.

Both tiers are in scope; A1-adaptive is the operationally dangerous expression and should be distinguished in reporting wherever telemetry permits. *Evaluation note:* fixed playbooks with branching satisfy neither tier unless branch selection is conditioned on environment state in a way that constitutes consequential choice; the boundary is whether the branch structure was fully enumerated in advance (policy execution) or generated/selected at runtime (autonomy). *v8 addition:* the human branching bound (Section 5.5) and exploration-entropy statistics (Section 5.3) provide behavioral evidence for A1 in trace-free environments — exploratory breadth beyond human baselines, unattributable to declared automation, is the graph-level signature of runtime selection. This strengthens A1 evaluability precisely where the autonomy-chain primitive is unavailable.

A2. Self-Propagation. S causes the creation of new operational instances or equivalent control footholds in remote contexts, where expansion is initiated by S's own operational logic rather than by an independent human operator. The foothold form is deliberately broad — process, container, service, scheduled task, credential implant, remote agent — because the class is defined by *acquired capability*, not artifact type. A propagator that implants only credentials and never drops a binary still propagates.

A3. Network-Level Propagation. Propagation crosses at least one meaningful network or administrative boundary: host-to-host, subnet-to-subnet, account-to-account, service-to-service, cluster-to-cluster, environment-to-environment. ANP is not a single-host construct. *Evaluation note:* "meaningful boundary" is environment-relative and must be declared in the conformance statement — in graph terms (Section 5.1), the declared boundary set is a partition over V, and A3 requires at least one observed edge crossing a partition cell.

A4. Propagation Dynamics (revised in v8). S's expansion through the environment exhibits statistically anomalous propagation dynamics relative to calibrated baselines. Formally, A4 is satisfied when the P-phase detector of Section 5 fires: the statistic vector (exploration-entropy gradient $\dot{H}_X$, weighted reach gain ΔW_R, distributional divergence D) exceeds calibrated stochastic bounds — by instantaneous spike (Section 5.4), by CUSUM accumulation (Section 6.1), or by surgical weighted-reach criteria (Section 6.3) — and the excess is not attributable to declared automation or ticketed human change (Section 5.5).

This replaces v6/v7's fixed context-count thresholds (≥N contexts in window T) with environment-calibrated stochastic bounds, eliminating the brittleness of hard host counts and unifying the epidemic/surgical split under one detector family: A4-Epidemic corresponds to spike detection on $\dot{H}_X$, A4-Surgical to ΔW_R and CUSUM criteria with corroborating anomaly. The legacy fixed thresholds are retained as the **fallback calibration profile** (Section 4.4) for environments unable to maintain the graph model. The minimum-activity floor carries forward: arbitrarily slow drift remains out of scope unless cumulative statistics (Section 6.1) cross calibrated bounds — unknown does not become signal merely by being slow, and slow does not become exempt merely by being patient.

A5. Structural Coherence. Footholds created by S preserve recognizable behavioral family resemblance across contexts. Exact copying is not required — polymorphism is expected — but the shape of behavior must identify the footholds as products

of one propagator logic. Coherence is anchored to a minimum feature rule: similarity in **at least two** of the following families, with thresholds stated in the conformance claim (Jaccard overlap over normalized feature tokens; graph-edit, spectral, or divergence bounds for topologies):

1. Process or execution graph topology
2. Tool or command-sequence overlap
3. Credential or identity reuse patterns
4. Timing or cadence regularity
5. Repeated control-plane modification patterns

v8 addition — predictive coherence: where phase structure is observable, A5 acquires a temporal dimension: the S-phase impact set must overlap the P-phase exploration set beyond random expectation (Section 5.6). This binds the phases to one propagator logic and is the framework's principal defense against false unification — attributing unrelated compromises to a single propagator — now formalized with an explicit null model rather than left to feature similarity alone.

A6. Runtime, Network-Native Operation. Propagation occurs through runtime communication or control paths: API calls, HTTP(S), SSH, SMB, RPC, remote orchestration actions, identity reuse, service-mesh calls, or similar. Pure build-time or supply-chain compromise without runtime spread is out of scope (see the ANP-S profile for supply-chain-*seeded* runtime propagation). Runtime operation is what makes the class observable in operational telemetry — and observability is a design requirement of the definition.

4.3 Exclusions and Boundary Cases

Explicitly outside the core model: purely human-operated intrusions with operator-chosen steps (fails A1); single-host malware (fails A2/A3); pure build-time compromise with no runtime spread (fails A6); benign orchestration, patching, scanning, autoscaling, and deployment — excluded both by the conjunction requirement and by declared-automation attribution (Sections 4.4, 5.5): activity attributable to declared, authorized automation is baseline, not signal.

Human-Guided Autonomous Propagation (in scope). A human sets objectives or boundary conditions; the system autonomously selects operational steps, targets, or methods. In scope when operational propagation satisfies A1–A6 — objective-level human involvement does not remove step-level autonomy. This is the near-term realistic case, treated as first-class.

Hybrid Seeding (in scope). S begins from a foothold created by other means — insider action, credential theft, supply-chain compromise — and subsequently satisfies A1–A6 at runtime. The defining question is whether subsequent spread is governed by S's own decision logic; seeding mechanism is provenance metadata, not a classification criterion.

Dormancy and windowing. A propagator satisfying the clauses in window W_1, dormant through W_2, resuming in W_3 is one ANP with an evidentiary gap, provided A5 coherence links the windows. Implementations should support cross-window linkage with decayed confidence. *v8 note:* dormancy between P-phase and S-phase is

expected adversary behavior, not an edge case — the predictive-coherence window (Section 5.6) must be calibrated longer than spike windows for precisely this reason.

4.4 Calibration Profiles

A4's stochastic bounds require baseline estimation. The procedure generalizes v7's velocity calibration:

1. **Select a calibration window** capturing normal operational variation, including deployment cycles and incident churn (30–90 days).
2. **Build the attribution baseline:** enumerate declared-automation contexts V_A and their characteristic graph signatures (topology templates, timing distributions, branch factors), and ticketed human-change activity V_H from change-management metadata. Declaration is an explicit inventory act; undeclared automation is, by design, indistinguishable from candidate signal — a deliberate organizational incentive to inventory automation.
3. **Estimate baseline statistics:** distributions of $\dot{H}_X$, ΔW_R, D, and branch factor B(t) during declared-automation activity and during human-attributed activity, conditioned on time-of-day/change-window where material; record means μ and deviations σ per statistic.
4. **Set detector parameters:** spike multiplier k (default 3), CUSUM reference value κ and decision interval h (Section 6.1), surgical weight criteria, predictive-coherence margin δ and window.
5. **Re-calibrate** on major changes in fleet composition, orchestration style, or architecture; preserve calibration history as conformance evidence.

Fallback profile (legacy v6/v7 thresholds, non-normative). Environments unable to maintain G(t) may operate fixed thresholds, accepting reduced robustness:

Environment tier	Epidemic guidance	Surgical floor
Small enterprise	≥3 new hosts in 10 min	≥2 contexts in 60 min + strong identity/control-plane anomaly
Mid-size enterprise	≥10 new contexts in 30 min	≥3 contexts in 90 min + strong coherence and anomaly support
Large enterprise / cloud	max(20 contexts, 1% of fleet) in 30 min	≥5 contexts in 2 hr + elevated coherence and control-plane signals

Conformance claims must state which profile (gradient or fallback) and which parameter values are deployed.

4.5 The Determination Procedure

A conforming implementation evaluates a candidate event stream as follows:

1. **Normalize** telemetry to the event schema (Section 7.1); maintain the environment graph G(t) and attribution sets V_A, V_H; attribute events to a candidate system S via coherence-linked correlation.
2. **Evaluate clauses independently:** each of A1–A6 receives a status (satisfied / not satisfied / indeterminate), an evidence set (specific events, statistics, primitive outputs), and a confidence value reflecting telemetry coverage.
3. **Classify with phase resolution (Section 5.8): ANP-forecast** when P-phase criteria hold (A4 fired, non-attributable, A1 behavioral indicators present) but no S-phase impact is yet observed; **ANP-confirmed** when all six clauses are satisfied including predictive coherence; **ANP-candidate** when no clause is refuted but one or more are indeterminate from coverage gaps; **not-ANP** when any clause is refuted.
4. **Report** the explained conjunction: per-clause status, evidence, confidence, phase state, applicable profile (core / ANP-N / ANP-S / ANP-E; gradient or fallback calibration), and calibration version in force.

The four-valued outcome is deliberate. Forecast is the operationally valuable state — it is where lead time lives — and candidate preserves the honest expressiveness of "consistent with ANP, autonomy unprovable" in coverage-limited environments.

---

5. The Seismic Propagation Model

This section formalizes the predictive core of v8. All logarithms are base 2 (entropy in bits).

5.1 Environment Model

Model the environment at time t as a directed multigraph:

G(t) = (V, E(t))

where V is the set of contexts (hosts, containers, accounts, services, clusters, tenants) and $E(t) \subseteq V \times V \times T$ the observed runtime reachability edges at time t, typed by $\tau \in T$ = {authentication transition, remote execution, network connectivity, control-plane privilege, service trust}. Edges carry timestamps and attribution metadata.

Two attribution subsets are maintained from the calibration inventory (Section 4.4):

- **$V_A \subseteq V$** — declared automation contexts (orchestrators, CI/CD, autoscalers, patch systems), with associated topology templates and timing signatures
- **$V_H \subseteq V$** — ticketed human-change contexts: activity linked to approved change-management records

Activity attributable to V_A signatures or V_H tickets is **baseline**; the detector statistics below are computed over residual (non-attributed) activity. Attribution is evidence-linked: an attribution decision cites the template or ticket it matched, and is itself reviewable.

Attribution maturity is an operational precondition, stated bluntly. The residual-statistics architecture assumes an accurate automation inventory, reliable

change-ticket linkage, and template fidelity — assumptions that are aspirational in many organizations. The framework's behavior under degraded attribution is therefore specified, not left implicit: (a) unattributable legitimate automation pollutes the residual and surfaces as candidate signal — by design, since the framework cannot and should not distinguish undeclared automation from autonomous intrusion (the organizational remedy is declaration, and candidate-band review is the safety valve); (b) sustained attribution leakage that is *shape-consistent* with declared templates raises residual volume but not divergence, and quantile-calibrated thresholds absorb it (a result confirmed empirically in Section 11.5: degraded-attribution storms produced no false-forecast elevation when leakage remained template-shaped); (c) the dangerous regime is *shape-anomalous* leakage — off-template human operations leaking into the residual at volume — which inflates false forecasts and must be bounded by calibration that includes representative leakage, per Section 4.4 step 3; and (d) calibration must exercise every detector regime the deployment will encounter (volume, reach-gain, divergence-supported windows) — degenerate calibration in which a statistic's baseline variance is near zero converts that statistic into a hair trigger, a failure mode identified during the Section 11.5 study and now a named conformance requirement. Deployments unable to meet a minimum attribution bar should run the fallback profile or ANP-N posture and treat forecast determinations as triage signals only.

5.2 Reach and Exploration: Two Distinct Quantities

v8 separates two quantities that earlier drafts of the entropy approach conflated, because they answer different questions and behave differently under attack:

Reachable set (potential). For a suspected origin v_0:

$$\mathbf{R(v_0, t) = \{ v \in V \mid \exists \text{ path } v_0 \to v \text{ in } G(t) \}}, \; r(t) = |R(v_0, t)|$$

R is primarily a property of the *environment* — it changes when privileges, rules, or topology change, regardless of which actor changed them. Quantities over R measure what a foothold *could* do. They drive response urgency and blast-radius forecasting (the propagation-potential lineage, Section 7.9), and they contribute the **weighted reach** statistic below — but they are not, by themselves, evidence of exploration.

Observed contact distribution (activity). Over window Δt ending at t, let $n_i(t)$ be the count of non-attributed transition events from the candidate system S contacting context i (authentication, connection, execution, control-plane action). Define the empirical exploration distribution:

$$\mathbf{q_i(t) = n_i(t) / \Sigma_j \, n_j(t)}$$

q measures what S *did*. P-phase detection keys on q; the environment graph supplies weights and reach.

Context weights. Each context i carries weight $w_i > 0$. Because ΔW_R carries surgical detection (Section 6.3), the weight model is the framework's most load-bearing quantitative assumption, and v8.1 specifies a principled default rather than leaving weights to hand-tuning. The default derivation is structural — computed from measurable graph properties, transferable across environments, and adversary-aware in

the specific sense that it approximates an attacker's value function rather than a defender's labeling effort:

w_i = (1 + outdeg_priv(i)) · (1 + ctl(i)) · s_i

where outdeg_priv(i) is context i's out-degree in G restricted to privilege and trust edges (what i can reach — lateral potential as measured, not as labeled), ctl(i) is a control-plane authority indicator or count (whether compromise of i permits modification of E(t) itself — IdP admins, orchestrators, CI/CD roots), and s_i is a sensitivity multiplier from declared data/impact classes, defaulting to 1 where no classification exists. The first two factors are computed from the same graph the framework already maintains; only s_i requires human input, and the model degrades gracefully without it. Hand-assigned weights remain permitted but must be declared in the conformance claim, and **weight sensitivity analysis is a required reported experiment** (Section 11.7): detection results must be shown stable under perturbation of the weight model, or the instability must be reported. The adversary-awareness argument: an attacker targeting value moves toward high-privilege, high-authority, high-sensitivity contexts almost by definition of "value" — a weight model derived from reach and authority therefore tracks the attacker's own objective function, which is what makes ΔW_R hard to avoid: evading the weighted statistic means avoiding the targets worth attacking.

5.3 The Detection Statistic Vector

No single scalar survives the full evasion space (Section 6 demonstrates this constructively). P-phase detection therefore computes a vector of three statistics over non-attributed activity, each catching what the others miss:

(1) Exploration entropy and its gradient (breadth).

$H_X(t) = -\Sigma_i\, q_i(t) \log_2 q_i(t)$

$\dot{H}_X(t_k) \approx [H_X(t_k) - H_X(t_{k-1})] / (t_k - t_{k-1})$

High H_X with growing support = broad, dispersed exploration — the epidemic P-phase signature. Interpretation requires care: entropy is maximal under *uniform* contact and **decreases under concentration**, so H_X alone is blind to focused, high-value targeting. That blindness is structural, not parametric, which is why the vector contains:

(2) Weighted reach gain (value capture).

$W_R(t) = \Sigma_{i \in R(v_0,t)}\, w_i$, $\Delta W_R(t_k) = W_R(t_k) - W_R(t_{k-1})$

ΔW_R registers expansion into *valuable* regions of the graph regardless of breadth: a propagator that touches only three contexts but acquires domain-admin reach produces a large ΔW_R with negligible entropy movement. This is the surgical-attack detector (Section 6.3), and it is deliberately a *gain*, not a distribution statistic — concentration cannot hide inside it.

(3) Distributional divergence (shape).

$D(t) = D_{KL}(q_t \,\|\, q_{base})$

the Kullback–Leibler divergence of the observed contact distribution from the calibrated baseline distribution for the matched activity class (declared-automation template during change windows; peer-group human baseline otherwise). D registers exploration whose *shape* deviates from what its volume profile claims to be — the mimicry detector (Section 6.2). Graph spectral divergence between observed and template topologies is an admissible alternative or supplement where distribution support is sparse; the conformance claim states the divergence functional used.

Each statistic is computed per candidate origin and per coherence-linked candidate system, with coverage annotations as for all primitives (Section 7.1).

5.4 Spike Detection (Epidemic P-Phase)

Baseline behavior of each statistic x ∈ {Ḣ_X, ΔW_R, D} is estimated during calibration as mean μ_x and standard deviation σ_x over declared-automation and human-attributed activity (Section 4.4). An **exploratory spike** fires when:

x(t) > μ_x + k·σ_x

with k a calibration constant (default 3), for any statistic in the vector, on non-attributed activity. This replaces hard host-count thresholds with environment-relative stochastic bounds: the same detector logic serves a 50-host enterprise and a 50,000-workload cloud estate, because the bounds are estimated from each environment's own baseline variation. Where baseline distributions are materially heavy-tailed, quantile bounds (99th/99.9th percentile) substitute for σ-multiples; the conformance claim states which.

5.5 The Human Branching Bound

Human operators — including skilled attackers operating interactively — exhibit constrained branching: bounded parallelism, serial attention, human-timescale interaction gaps. Let B(t) be the branch factor of non-attributed activity (distinct new targets contacted per source context per interval), and estimate the human baseline E[B]_hum and deviation σ_B from interactive-session telemetry, conditioned on peer group and time-of-day where material. The **exploratory autonomy condition** fires when:

B(t) > E[B]_hum + k_B·σ_B

AND no declared-automation signature or change ticket attributes the activity. This is behavioral evidence for clause A1 in trace-free environments: branching beyond the human envelope, unattributable to inventoried automation, indicates runtime target generation. It is corroborating evidence, not proof — a human launching an undeclared script exceeds the bound — which is precisely why determination requires the conjunction and why undeclared automation lands in candidate review rather than automatic high-band response. The bound's formalization parallels the spike rule deliberately: one alarm grammar across all detector statistics.

5.6 Predictive Coherence

Let **P** ⊆ V be the set of contexts contacted during a detected P-phase, and **S_I** ⊆ V the set of contexts impacted during a subsequent S-phase (Section 5.7). Define the overlap coefficient:

Overlap(P, S_I) = |P ∩ S_I| / |S_I|

Under the null model — S_I drawn without regard to exploration — the expected overlap is the weighted mass of P in the candidate population: with uniform weights, **E_rand = |P| / |V*|, where V* is the reachable candidate set; with weights, E_rand = Σ_{i∈P} w_i / Σ_{i∈V} w_i. *Significance is assessed against the hypergeometric tail (drawing |S_I| contexts from V* with |P| marked), at a documented significance level.** Predictive coherence holds when:

Overlap(P, S_I) > E_rand + δ

for calibrated margin δ with hypergeometric significance. This condition does three jobs: it binds the phases to one propagator logic (the temporal dimension of A5); it prevents false clustering of unrelated anomalies into a fabricated campaign; and it operationalizes the framework's falsifiability — if observed propagators' impact sets do not overlap their exploration sets beyond chance, the predictive model is wrong and Section 11's core experiment will say so. The coherence window must be calibrated longer than spike windows, because P→S dormancy is expected adversary behavior (Section 4.3).

5.7 Phase Definitions

P-phase (exploratory entropy expansion). An interval of non-attributed activity in which the detection vector fires — by spike (5.4), by cumulative criteria (6.1), or by surgical criteria (6.3) — optionally corroborated by the branching bound (5.5), control-plane widening, and identity-graph expansion. The P-phase is the propagator *learning*: enumerating, probing, authenticating, traversing.

S-phase (structured impact). An interval exhibiting structural impact criteria: persistence establishment, exfiltration, entrenchment (privilege consolidation, defense degradation, telemetry tampering), or disruption. The S-phase is the propagator *spending* what it learned. S-phase detection reuses existing, mature detection content (EDR persistence analytics, DLP, integrity monitoring); v8's contribution is not better S-phase detection but the claim that the P-phase *predicts* it — in time and in location.

5.8 Phase-Resolved Classification

The classification rule of the determination procedure (Section 4.5), stated formally:

ANP-forecast iff, over the evaluation window: 1. The P-phase detector fires (spike, CUSUM, or surgical criteria) on non-attributed activity; and 2. Attribution against V_A signatures and V_H tickets fails (the excess is unexplained); and 3. A1 indicators are present (autonomy-chain evidence where traces exist; branching bound and adaptive-ordering evidence otherwise); and 4. No clause is refuted.

ANP-confirmed iff additionally: 5. An S-phase satisfying structural impact criteria is observed; and 6. Predictive coherence (5.6) holds between the P-phase exploration set and the S-phase impact set; and 7. All six clauses of Section 4.2 are satisfied with the evidence assembled across phases.

The split is the load-bearing correction of v8's determination logic: a rule requiring S-phase observation before any classification would reduce the framework to post-hoc confirmation and forfeit the early-warning value that justifies the predictive layer. Forecast determinations trigger reversible P-band response (Section 8.3); confirmed determinations unlock the full response ladder. Forecast determinations that never confirm are not failures — they are either successful early containment (the propagator was stopped before impact), false alarms (measured, Section 11.2), or stealth beyond the model's reach (Section 6.5); distinguishing these outcomes is itself part of the validation program.

5.9 Minimum Telemetry

The seismic layer operates on four feeds: **identity transitions, network adjacency changes, control-plane modifications, and change-ticket metadata.** Autonomy traces (ANP-E) strengthen A1 evidence but are optional. This is a deliberate reduction from the rich-telemetry assumptions of earlier versions: the graph statistics are computable from logs most mature environments already collect, which materially widens the framework's deployable surface and strengthens the intermediate tier of the availability matrix (Section 7.10).

5.10 The Exploration Lower Bound

The seismic model's foundation is the claim that adaptive propagation obliges exploration. v8.1 formalizes that claim, converting it from analogy to bounded consequence.

Setting. Let the environment's relevant state be $\theta \in \Theta$, unknown to the propagator beyond a prior π. Suppose achieving the propagator's objective set Ω with success probability $\geq 1 - \varepsilon$ requires correctly determining a decision variable $d(\theta) \in D$ — which of M candidate propagation plans (target orderings, credential paths, pivot sequences) is viable — where d is not constant under π. Let $H_\pi(d)$ denote the propagator's prior uncertainty about d in bits; under a uniform prior over M viable-plan hypotheses, $H_\pi(d) = \log_2 M$.

Proposition 1 (Exploration lower bound). Any policy that selects a plan achieving success probability $\geq 1 - \varepsilon$ must acquire mutual information about d satisfying, by Fano's inequality [21]:

$$\mathbf{I(d\ ;\ observations) \geq H_\pi(d) - h(\varepsilon) - \varepsilon \cdot \log_2(|D| - 1)}$$

where $h(\varepsilon)$ is the binary entropy function. If all of the propagator's information channels are **in-band** — each probe, query, authentication attempt, or traversal is an event in the defender's telemetry domain — and each such event yields at most c bits of information about d (c bounded by the response alphabet of the probed interface), then the propagator must generate at least

$$n \geq [H_\pi(d) - h(\varepsilon) - \varepsilon \cdot \log_2(|D| - 1)] / c$$

defender-observable events, and these events must be distributed across the support of the propagator's uncertainty — contexts about which it lacks information — rather than confined to contexts it already knows.

Proof sketch. The first inequality is Fano's inequality applied to the estimation of d from the propagator's observation history; any decision rule with error probability ≤ ε requires the stated mutual information. The second follows from the data-processing inequality and channel additivity: n events each carrying at most c bits yield at most nc bits of mutual information. The support condition follows because observations of already-determined components of θ have zero conditional information about d. □

Corollary 1 (Stealth is priced in priors). Out-of-band knowledge — exfiltrated IaC repositories, purchased network maps, insider briefings — reduces $H_\pi(d)$ and therefore reduces the bound, reaching zero only when the prior fully determines d. This is the formal content of the stealth limitation (Section 6.5): the P-wave is not unconditionally obligatory; it is obligatory *in proportion to the propagator's residual uncertainty*, and an adversary who eliminates it has paid for the elimination through an out-of-band acquisition channel with its own costs, risks, and detection surfaces (which lie outside this framework's scope and inside the broader defense-in-depth posture).

Corollary 2 (Adaptivity premium). A1-adaptive behavior — replanning against discovered state, retargeting around containment — requires *re-estimating* d as θ changes, renewing the information requirement each time the environment or the defender's response invalidates prior knowledge. Adaptive propagators therefore face the bound repeatedly; static playbooks face it once (or never, with sufficient priors) but forfeit adaptation. This is the trade-off of Section 6.5 stated exactly: the bound prices autonomy's defining advantage in observable events.

What the proposition does and does not establish. It establishes that in-band adaptive propagation under uncertainty must generate observable exploration events at a rate bounded below by information requirements — the *existence* of the P-wave under stated assumptions. It does **not** establish that any particular statistic detects those events against baseline noise: the events may individually resemble benign activity (low per-event anomaly), arrive slowly (the CUSUM regime), or hide in attribution leakage. Detection inevitability would additionally require assumptions about baseline separability that are empirical, not information-theoretic — which is why Proposition 1 grounds the model while RQ5 and the validation program remain the arbiters of practical value. The open problem of tightening c, characterizing $H_\pi(d)$ for realistic objective classes, and connecting the event-count bound to specific statistic exceedances is Research Agenda item 5.

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6. Adversarial Stress Tests of the Seismic Model

The model is evaluated constructively against five adversaries. Each attack defines the statistic that defeats it — this section is why Section 5.3 is a vector.

6.1 Slow-and-Low: CUSUM Accumulation

Attack. Expand exploration slowly enough that every per-interval statistic stays under the k·σ spike bound.

Mitigation. Per-interval thresholds are the wrong instrument for small persistent shifts; the right one is the cumulative sum control chart [18]. For statistic x with baseline mean μ_x, define:

$$\mathbf{C_k = \max(0,\ C_{k-1} + (x(t_k) - \mu_x - \kappa))},\ C_0 = 0$$

with reference value κ (the slack: shifts smaller than κ per interval are tolerated) and decision interval h: alarm when **C_k > h**. CUSUM accumulates *persistent* positive deviation while random fluctuation self-cancels through the max(0,·) reset. A patient propagator must keep its expected per-interval excess below κ *indefinitely* — and κ is set so that staying below it means exploring at a rate whose cumulative reach, over the same horizon, stays within declared-baseline drift. Slow evasion thus becomes a real constraint on the adversary's objective, not a free pass. (A note on a tempting simplification: the integral $\int \dot{H}_X\, dt$ telescopes to $H_X(t_n) - H_X(t_0)$ — net change, which says nothing about *when* deviation occurred and resets to zero if exploration ends where it began in distribution. CUSUM on increments against baseline drift is the formally correct accumulator, which is why v8 specifies it rather than the integral.)

6.2 Automation Mimicry: Shape Divergence

Attack. Expand along declared automation paths, during change windows, using maintenance identities — hide volume inside authorized volume.

Mitigation. Compare exploratory *shape*, not volume. Declared automation is template-driven: its contact distributions and topologies are stochastic in scale (autoscalers vary in instance count) but conservative in shape (the same relative structure, the same edge types, the same ordering grammar). Autonomous exploration is probabilistic and branching — conditioned on discovered state, it deviates from any fixed template as it pursues value. The divergence statistic $D(t) = D_KL(q_t \,\|\, q_template)$ — or graph spectral divergence $D_spec(G_obs, G_template)$ where support is sparse — registers shape deviation at matched volume. The deeper property: to defeat D, the mimic must constrain its exploration to the template's distribution, which means *not exploring* where the template does not go — mimicry that defeats the detector also forfeits the exploration the propagator needed. The defense converts evasion into a forgery problem priced in operational capability.

6.3 Surgical Targeting: Weighted Reach Gain

Attack. Expand to only two or three high-value contexts; produce negligible breadth.

Mitigation. This adversary defeats entropy *by design* — concentration lowers H_X; no spike occurs, and no parameterization of an entropy threshold fixes that, because the blindness is structural. The vector's second component exists for this case: ΔW_R registers acquisition of weighted reach regardless of breadth. Two contexts that confer control-plane authority over a tenant move W_R more than two hundred workstations. Surgical criteria: ΔW_R exceeding its calibrated bound (spike or CUSUM form),

with corroborating identity-walk or control-plane anomaly and two-family coherence — the v8 form of A4-Surgical. (Earlier drafts of the entropy approach asserted that weight concentration produces an entropy spike; it produces the opposite, and v8's correction is to stop asking entropy to do this job.)

6.4 Patch-Storm False Positive: Attribution Veto

Attack-shaped event. A massive legitimate scaling or patch event produces genuine entropy expansion.

Mitigation. Attribution before alarm: change-ticket correlation (V_H), declared-automation signature matching (V_A), and known topology-template replication. Entropy rises, but the rise is attributed — the statistics are computed over *residual* activity (Section 5.1), and an attributed storm leaves no residual. Where attribution is partial (a storm plus something inside it — the noise-cloaking case), the divergence statistic operates on the residual against the storm template, and cross-tenant correlation distinguishes local churn from a propagating class. The benign-automation veto of the confidence bands (Section 8.2) backstops the statistics: a high-confidence declared-automation match caps the band pending review.

6.5 Fully Stealth AI: The Honest Limitation

Attack. Minimal in-band exploration; near-linear path to objectives, indistinguishable from a skilled human operator's footprint.

Result. No detectable P-wave. v8 does not detect fully linear stealth propagation, and this paper declines to pretend otherwise. Two observations bound the damage, and Proposition 1 (Section 5.10) now grounds both formally: the P-wave is obligatory in proportion to the propagator's residual uncertainty (Corollary 1), and adaptivity renews the obligation (Corollary 2). First, the limitation is broader than "perfectly stealthy AI": exploration can be skipped whenever environmental knowledge is acquired *out-of-band* — exfiltrated IaC repositories, purchased network maps, insider knowledge — so P-phase detection covers in-band learning only, and the validation program measures incidence of out-of-band-primed attacks as a bound on coverage (Section 11.5). Second, the constraint imposed on the adversary is real and compounding: a propagator that forgoes runtime exploration forfeits exactly the adaptive advantage that distinguishes the class — it cannot react to discovered state it never discovers, cannot retarget around containment it never probes, and degrades toward a static playbook detectable by conventional means and brittle against environment drift. The trade-off is imposed, priced, and acceptable: v8 makes autonomous *search* expensive; an adversary who abandons search has been partially defeated before detection is even attempted.

6.6 Summary Table

Adversary	Defeats	Caught by	Residual cost imposed
Slow-and-low	Spike bounds	CUSUM accumulation	Exploration rate capped below baseline drift
Automation mimicry	Volume thresholds	KL/spectral divergence vs. template	Exploration confined to template support
Surgical targeting	Entropy statistics	Weighted reach gain ΔW_R	High-value moves are exactly the weighted ones
Patch-storm (benign)	— (false-positive risk)	Attribution veto + residual analysis	—
Fully stealth / out-of-band-primed	Entire P-phase detector	Not detected (stated limitation)	Forfeits adaptive search advantage

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7. Detection Primitives

The primitive library carries forward from v7 with two additions (P8, P9) operationalizing the seismic layer. Primitives are deliberately smaller than detections: they compute evidence; the determination procedure and confidence bands compose evidence into claims. Each primitive can be validated, ablated, and replaced independently (RQ2).

7.1 Interface Conventions

Each primitive declares an input schema, output schema, computation mode (streaming/batch/hybrid), and failure/degradation mode. Degradation must be visible — a primitive on impoverished input says so in its coverage field, never silently emitting a confident score.

```
Event {
  timestamp: DateTime,
  source_context: ContextID,
  target_context: ContextID,
  identity: IdentityID,
  event_type: EventType,
 attribution: Optional<AttributionRef>,  // v8: V_A template or V_H ticket match
  attributes: Map<String, Value>
}

PrimitiveScore {
  primitive_id: String,
  score: Float[0,1],
  confidence: Float[0,1],
```

```
  coverage: Float[0,1],
  evidence: List[EventRef],   // normative: scores must cite events
  notes: String
}
```

7.2 Primitive 1: Propagation Graph Score

`propagation_graph(events, dt) -> PrimitiveScore` — **Clauses A2, A3, A4. Hybrid.**

Build the directed foothold graph; compute longest propagation chain, mean branch factor from newly affected nodes, descendant ratio (proportion of contexts appearing only after being reached by another — high values indicate spread structure rather than coincident compromise), and temporal compression. Normalize against baselines → s_pg. *Failure:* degrades without remote-execution/foothold visibility; descendant ratio is most robust under partial coverage and should dominate when coverage < 0.5.

7.3 Primitive 2: Flow Fan-Out Score

`flow_fanout(flows, dt) -> PrimitiveScore` — **Clauses A3, A4 (supporting). Streaming.**

Per source context: distinct destination count, distinct prefix count, distinct service count, failed/incomplete connection ratio (probing residue). Compare to baseline; convert deviations to tail-probability scores → s_ff. *Failure:* usable in metadata-only environments (the ANP-N backbone) with reduced semantic specificity — sees spread shape, not content.

7.4 Primitive 3: Identity Walk Anomaly

`identity_walk(auth_events, dt) -> PrimitiveScore` — **Clauses A2, A3, A5 (identity family). Hybrid.**

Model each identity's historical resource/role paths; compare the window's path against the identity's own history and a peer group. Low path likelihood, unusual privilege transitions, expansion into new resource classes, compressed timing → s_iw. Peer-group comparison is required — it controls for legitimate role changes, the dominant benign cause of self-history deviation. *Failure:* strongly degraded under incomplete identity telemetry; emits low coverage, not low score.

7.5 Primitive 4: Autonomy Chain Score

`autonomy_chain(trace) -> (PrimitiveScore, PrimitiveScore)` — **Clause A1 (primary), intent. Batch.**

Where prompt/tool/response traces exist (ANP-E): loop depth, prompt→tool→prompt cycles, self-initiated steps, tool diversity, dependency of later steps on intermediate results absent human input → s_auto. Separately, estimate exploit-oriented intent against a documented action taxonomy (reconnaissance, credential access, discovery, lateral-movement planning, execution preparation, persistence, impact) → s_intent.

False-positive note: legitimate agents produce structurally similar traces; Primitive 4 never drives strong containment alone (bands enforce this). *Failure:* absent traces, autonomy is inferred weakly from behavioral sequences — in v8, principally via P8's branching bound and adaptive-ordering evidence — with coverage and confidence marked accordingly.

7.6 Primitive 5: Control-Plane Anomaly

`control_plane_anomaly(control_events, dt) -> PrimitiveScore` — **Clauses A2, A3, A5 (control-plane family), A6. Streaming.**

Identify actions that materially widen reach: broader ingress, new privileges, new peering, new reachable execution contexts. Score sequences by novelty against the declared baseline, breadth of widening, and proximity to other suspicious signals → s_cp. *Rationale:* cloud-native propagation is frequently configuration propagation — the propagator moves permissions, then walks through the doors it opened; in graph terms, control-plane widening is *direct manipulation of E(t)*, which is why s_cp is also a leading P-phase indicator. *Failure:* absent without control-plane logs.

7.7 Primitive 6: Cross-Tenant Correlation

`cross_tenant_correlation(signals) -> PrimitiveScore` — **A5 (cross-environment coherence); systemic awareness. Batch.**

Exchange format (minimum): time bucket, primitive IDs, scores, confidence, tenant class, region, telemetry profile, coarse context tags. Align signals in time; compare feature vectors; measure how many unrelated environments show unusually similar anomaly structure in a narrow window → s_ct. *v8 note:* P-phase statistics summarize well — entropy gradients and divergence scores are low-dimensional and non-identifying, making the seismic layer naturally federable. *Threat note:* the correlation system is itself a target; injection, suppression, replay, and poisoning are in its threat model.

7.8 Primitive 7: Propagation Potential Score

`propagation_potential(graph, context, dt) -> PrimitiveScore` — **Response prioritization only. Batch.**

Constrained reachability from a candidate foothold over network reachability, identity privileges, execution pathways, service trust, and control-plane permissions; weight the reachable set by privilege breadth, adjacency type, movement likelihood, and rare-permission dependence → s_pp. In v8 terms, s_pp is the normalized form of weighted reach W_R computed *prospectively*. **Role distinction preserved:** s_pp is excluded from classification — what a foothold could do is not evidence of what the system is. It governs response urgency only. (Its differential form ΔW_R, computed over *observed* reach acquisition, is classification evidence — the distinction is potential versus acquired, and v8 makes it precise.)

7.9 Primitive 8 (new): Entropy Gradient Score

`entropy_gradient(events, graph, dt) -> PrimitiveScore` — **Clause A4 (primary), A1 (behavioral support). Streaming.**

Computes the Section 5.3 statistic vector over non-attributed activity: exploration entropy H_X and gradient Ḣ_X over the observed contact distribution; weighted reach gain ΔW_R; divergence D from the matched baseline template; branch factor B(t) against the human bound; CUSUM accumulators per statistic. Output score reflects the strongest calibrated exceedance across the vector; evidence cites the contributing events and the statistic(s) that fired; notes identify the detection mode (spike / CUSUM / surgical). *Failure:* requires the environment graph and attribution inventory; degrades to flow-fan-out-equivalent breadth signals without weights, and to spike-only detection without baseline history. This primitive is the P-phase detector; it is the v8 operationalization of A4 and the behavioral channel for A1.

7.10 Primitive 9 (new): Predictive Coherence Score

`predictive_coherence(p_set, s_events, graph) -> PrimitiveScore` — **Clause A5 (temporal dimension); forecast→confirmed transition. Batch.**

Given a recorded P-phase exploration set and candidate S-phase impact events: compute Overlap(P, S_I), the null expectation E_rand under the weighted hypergeometric model, and tail significance; score reflects exceedance over E_rand + δ. Evidence cites the overlapping contexts and the phase-linking events. *Failure:* undefined until an S-phase candidate exists (this primitive only participates in confirmation, never forecast); degrades when the P-set is stale relative to the calibrated coherence window.

7.11 Composition Semantics

1. **Plane independence.** For confidence-band purposes, two primitives are independent iff they consume disjoint telemetry planes. P8 spans identity/network/control-plane feeds: it counts as one plane (its dominant feed) for independence accounting, declared per deployment in the conformance claim.
2. **Coverage-weighted aggregation.** Aggregates weight by coverage; a high score on 10% coverage must not dominate a moderate score on 95%. Aggregation functions are documented and monotone in score and coverage.
3. **Evidence union.** Composed claims cite the union of constituent evidence sets; no composition step discards evidence references.

7.12 Primitive Availability by Environment

Primitive	Rich telemetry	Intermediate	Metadata-only
P1 Propagation graph	Strong	Moderate	Partial, coarse
P2 Flow fan-out	Strong	Strong	Strong
P3 Identity walk	Strong	Moderate	Not available

Primitive	Rich telemetry	Intermediate	Metadata-only
P4 Autonomy chain	Strong	Weak	Not available
P5 Control-plane	Strong	Moderate	Not available
P6 Cross-tenant	Strong	Strong	Strong
P7 Propagation potential	Strong	Moderate	Partial, coarse
P8 Entropy gradient	**Strong**	**Strong**	**Partial (breadth-only)**
P9 Predictive coherence	**Strong**	**Moderate**	**Partial (flow-level sets)**

P8's strength at the intermediate tier is the practical payoff of Section 5.9's minimum-telemetry design: most mature environments can run the P-phase detector today.

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8. Response Primitives and Phase-Keyed Containment

If the propagator operates at machine speed, part of the defensive loop must operate at machine speed — bounded, reversible, governed. v8 adds phase-keying: the P/S structure tells the defender not only *that* to act but *when acting is cheap*.

8.1 Response Primitive Library

ID	Primitive	Trigger basis	Action	Reversibility	Authorization scope
RP1	Identity quarantine	Medium/high confidence + strong identity signals	Revoke sessions, reduce privileges, step-up auth	High	Identity provider
RP2	Context segmentation	High confidence, or medium + high s_pp	Tighten network policy around host/subnet/cluster	Medium	Network policy authority
RP3	Session termination	Medium/high confidence + active suspect sessions	Kill or block sessions on suspected path	High	Session control authority

ID	Primitive	Trigger basis	Action	Reversibility	Authorization scope
RP4	Control-plane freeze	High confidence + strong control-plane anomaly	Temporarily block sensitive route/privilege/topology changes	Medium	Cloud/orchestration admin
RP5	Execution hold	Medium confidence + suspicious automation/pipeline	Pause job/task/deployment pending verification	High	Orchestration/CI-CD authority
RP6	Evidence sealing	Any medium/high confidence event	Commit event bundle, preserve logs, tamper-evident record	High	Security operations
RP7	Recovery and rollback	Post-false-positive or resolved incident	Restore access, thaw control plane, remove segmentation	High	Same as original response

RP6 is unconditional at medium confidence and above: evidence preservation precedes and outlasts every other action — incident response, predictive-coherence confirmation (P9 needs the sealed P-set), and validation all depend on it, and a capable propagator's first defensive move is telemetry destruction.

8.2 Confidence Bands

Band	Criteria
Low	Any single primitive score > 0.6 with moderate coverage but insufficient cross-plane support
Medium	≥2 *independent* primitives > 0.7, OR one primitive > 0.8 plus strong contextual support
High	≥3 independent primitives > 0.8, with at least one from {P1, P3, P4, P5, P8}, and no strong benign-automation contradiction

Thresholds are non-normative defaults; the invariant is that stronger containment requires multi-plane corroboration. The benign-automation veto: a high-confidence match against the declared inventory caps the band at Medium pending review, regardless of scores.

8.3 Phase-Keyed Confidence-to-Action Mapping

Phase state	Confidence	Recommended action
—	Low	Enrich, label, preserve evidence, monitor; no automatic disruption
ANP-forecast (P-wave)	Medium	Reversible restrictions on the exploration's identity and path surface: RP1 step-up/session revocation, RP5 execution hold, RP6 sealing; targeted telemetry enrichment on forecast next-hop contexts (s_pp-prioritized); alert with evidence
ANP-forecast (P-wave)	High	Above, plus RP2 local segmentation around the exploration frontier and RP4 freeze on the specific control-plane surfaces being widened — still fully reversible; mandatory review clock starts
ANP-confirmed (S-wave)	Medium/High	Full ladder: RP1 quarantine, RP2 broad segmentation, RP4 control-plane freeze, RP3 termination; review and rollback path mandatory

The economics of the phase split: P-band actions are cheap *because* they are early — restricting an identity mid-exploration disrupts no production workload; the same restriction post-entrenchment fights established persistence. Lead time is not only warning; it is the period during which containment is reversible, narrow, and low-collateral. This is the operational content of RQ5's blast-radius arm.

8.4 Human-in-the-Loop Boundaries

- Low confidence never causes disruption automatically.
- Medium confidence may trigger reversible controls automatically.
- High confidence may trigger pre-authorized containment where the blast radius of inaction exceeds the blast radius of temporary disruption — pre-authorization being an explicit, recorded organizational decision, not a vendor default.
- All medium and high actions generate review obligations with named reviewers and deadlines, and rollback criteria stated at action time.
- Forecast-band actions carry a *forecast expiry*: if no S-phase or further P-phase evidence emerges within the calibrated coherence window, restrictions auto-expire through RP7 and the episode is logged as an unconfirmed forecast (a measured quantity, Section 11.2).

8.5 Rollback and Recovery

Every disruptive primitive ships with a paired recovery path: quarantine supports evidence-gated restoration; freeze has a thaw procedure with logging; segmentation carries timed expiration or review gates. The design target is not zero false positives; it is cheap, fast, auditable recovery from them — a framework optimized to never be wrong is too conservative to be useful.

8.6 The Economics of Autonomous Containment

Let C_fp be the expected cost of a false-positive response (disruption × duration × scope) and C_fn the expected cost of undetected propagation per unit time (spread rate × per-context damage). Reversibility engineering reduces C_fp; machine-speed response reduces the time term of C_fn; and the phase split reduces *both simultaneously* — P-band actions have intrinsically small scope (the exploration surface, not the estate) while buying the largest time savings (pre-entrenchment). The open empirical question is where the automation crossover sits: at what confidence, in which environments, does automated reversible containment dominate human-gated response? The bands and phase-keying are the parameterization that lets organizations position themselves on that curve deliberately.

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9. Telemetry Integrity, Variants, Governance, and Ethics

9.1 Telemetry Poisoning

A sophisticated propagator may delete, delay, forge, suppress, replay, or distort the telemetry feeding the primitives — for an autonomous system, attacking the sensors is just another path choice. Implementations should prefer: append-only or tamper-evident event stores; signed logs where platform support exists; redundant telemetry across planes (corroboration doubles as integrity checking); confidence penalties on unexpected coverage collapse; and **sensor silence as an event** — an unexplained drop in expected telemetry is a scored observation, never a neutral condition. In a framework whose claims are coverage-weighted, an adversary who silences sensors

can manufacture indeterminacy; treating silence as signal closes the loop. *v8 addition:* the attribution inventory is now also a poisoning target — forging a declared-automation signature or a change ticket converts signal into baseline. Attribution sources (template stores, change-management systems) therefore inherit the same integrity requirements as event stores, and attribution decisions are evidence-linked and reviewable (Section 5.1).

9.2 Model Poisoning and Model Integrity

Where the propagator's planning depends on an LLM, planner, or policy model, that component is an attack surface in both directions. Defender-side: autonomy-trace evidence is *potentially manipulated testimony*, corroborated by behavioral planes before high-band determinations. Symmetrically, defensive reasoning models (RQ3) inherit the same integrity threat model; their inputs are adversarial by default. Model integrity is part of the evidence problem, not a separate concern.

9.3 Evidence and Proof

Prefer local computation and summarized outputs; where an environment can compute a primitive score on-site and emit only the score and committed evidence references, that is preferable to exporting raw traces. High-severity determinations should be *provable rather than asserted* — signed event chains and Merkle-linked logs make downstream trust independent of vendor claims. This matters doubly for a framework that expects to be wrong sometimes: disputed determinations are settled by evidence inspection, and evidence must survive both the incident and the propagator's interest in destroying it (RP6).

9.4 ANP-N: The Network-Observable Variant

In environments limited to flows, routing data, or aggregate DNS, full determination is impossible — and a framework demanding impossible telemetry will be ignored by most real networks. **ANP-N** captures propagation-relevant behavior from metadata alone.

Can satisfy: A3 (strongly, via flow analysis); A4 (P8 in breadth-only degraded mode: contact-distribution entropy and fan-out are flow-computable); A5 (partially: repeated spread patterns as similar flow shapes or destination classes; P9 at flow-level sets). **Cannot satisfy:** A1 (autonomy unprovable from metadata); A2 (limited foothold visibility); A5 in full (no process graphs or tool sequences).

ANP-N determinations are *candidate-class* or *forecast-class* by construction: early warning, coarse detection, systemic analysis — inputs to escalation and cross-tenant correlation, not substitutes for full determination. *v8 note:* the seismic layer strengthens ANP-N materially — breadth-only P-phase detection plus cross-tenant correlation of entropy signatures gives metadata-limited environments a predictive capability they previously lacked entirely.

9.5 Privacy-Preserving Cross-Tenant Correlation

Exchange summarized signals, never raw traces: coarse time-bucketed feature vectors, hashed or bucketed anomaly indicators, secure aggregation of primitive outputs, federated computation of local features. The architectural principle precedes the cryptography: **correlation must not depend on exporting raw customer-identifiable content where a summarized signal suffices.** P-phase statistics are naturally low-dimensional and non-identifying — entropy gradients and divergence scores summarize without exposing — making the seismic layer the most privacy-compatible component of the framework. Minimality is simultaneously a privacy control, an attack-surface control, and an adoption control.

9.6 Governance, Conformance, and Profiles

ANP is specified as a versioned standard: published core specification under semantic versioning; public change-proposal process; a review group spanning operator, platform, and governance perspectives; higher ratification thresholds for core-clause changes than for clarifications; documented deprecation preserving cross-version comparability.

Conformance suite. Three case classes: **positive** (clear ANP behavior, multi-plane evidence, both phases), **negative** (legitimate orchestration/patching that must not classify), **ambiguous** (slow, surgical, or mimicry-oriented propagation stressing calibration). v8 conformance claims additionally state: calibration profile (gradient/fallback), parameter values (k, κ, h, δ, weights), divergence functional, independence declarations, and phase-window calibrations.

Extension profiles: ANP-N (metadata-limited, Section 9.4); **ANP-S** (supply-chain-seeded runtime propagation, with build-time provenance); **ANP-E** (embedded model traces; Primitive 4 at full strength).

9.7 Ethical Boundaries

ANP exists for detection, response, architecture, and systemic risk awareness — not generalized surveillance, hidden payload inspection, or opaque traffic discrimination. Minimum normative posture:

- ANP detection must not be repurposed into unrelated user-profiling systems
- Cross-tenant correlation must not be used to infer competitive intelligence about participants
- Response primitives must be logged, reviewable, and auditable
- Sensitive telemetry access must be role-limited and recorded
- Deception systems must remain within environments the defender legitimately controls

Implemented through audit logging, separation of duties, access control, review requirements, and retention limits — controls, not aspirations. A detection framework with privileged visibility into identity and control planes is itself dual-use, and this paper treats that as a design constraint: the privacy-preserving correlation architecture, the summarized-evidence preference, and the minimal-telemetry seismic layer

all exist so that defensive value does not require surveillance-grade collection. *v8 note:* the human branching bound (Section 5.5) profiles *aggregate behavioral envelopes*, not individuals; implementations must compute it over peer-group statistics with the same role-limited access as identity telemetry, and the bound must never be repurposed for workforce monitoring.

9.8 Deception and Active Defense

Because ANP includes a reasoning component, it can be misled as well as detected. Honey identities and service paths reveal credential abuse and opportunistic branching; honey prompts and documents redirect reasoning toward instrumented dead ends; planner exhaustion traps consume search effort on low-payoff exploration. *v8 addition:* deception has a natural synergy with the seismic model — honey resources are high-weight-appearing, zero-legitimate-traffic contexts, so exploration touching them produces clean P-phase evidence with near-zero false-positive risk; a well-placed honey layer effectively raises the P-wave's signal-to-noise ratio. Deception requires stronger guardrails than monitoring: evaluate collateral impact on legitimate users, compromise risk of the deception infrastructure, and capability leakage to the adversary. Deploy first in isolated, instrumented environments; remain bounded by Section 9.7.

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10. Worked Vignettes

Three synthetic scenarios — none describes a real incident — evaluated through the v8 pipeline with phase annotations. Together they exercise the three conformance classes, the three detection modes (spike, attribution-veto, surgical/CUSUM), and three of the four determination outcomes.

10.1 Vignette A — Epidemic Cloud Propagation (positive; spike mode)

Scenario. Mid-size SaaS environment. Over 24 minutes: a CI runner emits an anomalous IAM-enumeration sequence (control-plane); a service account authenticates to 11 never-before-touched services in privilege-ascending order inconsistent with its peer group (identity); new containers appear in three clusters, each within 90 seconds of the credential that reached it; each instance repeats a recognizable sequence — enumerate secrets, query metadata, open outbound connections to two new prefixes; fan-out from each context begins before the prior context's activity completes.

Seismic evaluation. Non-attributed contact distribution support grows from 3 to 26 contexts; $\dot{H}_X$ spikes at 5.8σ above declared-automation baseline (P8, spike mode); B(t) = 7 against E[B]_hum + 3σ_B ≈ 2.6 (branching bound fires); ΔW_R jumps on acquisition of cluster-admin-adjacent reach; D(t) high against all automation templates; no V_A signature or V_H ticket matches. **ANP-forecast declared at t+9 min** — P-band high: RP6 sealing, RP1 quarantine of the walking service account, RP2 segmentation around the three-cluster exploration frontier, RP4 freeze on the IAM surfaces being widened. **S-phase attempt observed at t+21 min** (persistence implants in two reached clusters); Overlap(P, S_I) = 1.0 against E_rand ≈ 0.04, hypergeometric

p □ 0.001 (P9). Clause evaluation: A1-adaptive (ordering tracks discovered privilege; branching bound), A2, A3, A4 (gradient profile), A5 (tool-sequence Jaccard 0.81 + control-plane pattern + predictive coherence), A6 — all satisfied.

Determination: ANP-confirmed (high). Measured lead time: **12 minutes** between forecast and first S-phase attempt — the quantity RQ5 aggregates. Blast radius at containment: 3 clusters segmented mid-exploration versus 9 forecast reachable (s_pp) at unchecked spread.

10.2 Vignette B — Benign Orchestration Storm (negative; attribution-veto mode)

Scenario. Retailer's Black-Friday autoscaling coincides with an emergency patch rollout: 240 new containers across 12 clusters in 30 minutes; a deployment account touches every cluster; near-identical process trees everywhere; massive flow fan-out.

Seismic evaluation. Raw entropy expansion is enormous — and almost entirely *attributed*: every foothold-creating action matches the declared autoscaler/pipeline signatures (V_A), parameterized by a recorded change ticket (V_H); the residual contact distribution after attribution is statistically empty; D(t) computed on the storm against its own template ≈ 0 (the storm *is* the template at scale); identity walk is the deployment account's normal peer-path at elevated volume (s_iw 0.22). A1 refuted on attribution: branch structure fully enumerated in the deployment manifest.

Determination: not-ANP. The explained non-determination — "high-volume activity fully attributed to declared automation, ticket #—, template match at 0.97" — is itself useful operational confirmation. Volume never classifies; this is the false-positive control the negative corpus class enforces, now mechanized by residual-statistics-after-attribution rather than by judgment.

10.3 Vignette C — Surgical Identity-Plane Propagation (ambiguous; surgical/CUSUM mode)

Scenario. Managed-services environment, intermediate telemetry, no autonomy traces. Over 9 days: four high-value contexts (backup server, IdP admin host, two finance services) acquire new scheduled tasks or service principals; each acquisition preceded by a short failed-then-successful authentication sequence from a different source context, always inside the local change window, always via a dormant-but-valid service identity; flow shapes between acquisitions nearly identical; inter-foothold gaps 31–62 hours.

Seismic evaluation. H_X barely moves — four contexts, concentrated weight: entropy is structurally blind here, as designed-for. ΔW_R tells the story: the four contexts carry 38% of estate weight (IdP admin alone: 19%), and cumulative ΔW_R crosses its CUSUM decision interval h on the third acquisition (day 6). D(t) on residual activity diverges from the change-window template (the mimic borrows the *schedule*, not the *shape*: its authentication grammar — fail-then-succeed from rotating sources — appears in no automation template). Branching stays within human bounds (the stealth-leaning case); s_iw = 0.84 on dormant-identity reactivation across four trust

domains; coherence holds on timing-cadence + identity-reuse families. A1: **indeterminate** — no traces; behavioral evidence (adaptive window-timing, value-ordered targeting) suggestive but below the bar at this coverage.

Determination: ANP-candidate (medium), forecast-equivalent response posture — RP1 step-up on the dormant identity, RP5 hold on pending scheduled tasks, RP6 sealing, enrichment on the two highest-ΔW_R forecast next-hops. The candidate class does its job: reversible mitigation and attention-routing without overclaiming, with a precise statement of which missing evidence (autonomy plane, or an observed S-phase with coherence) would upgrade or refute. *v8 versus v7 on this vignette:* v7's fixed surgical floor caught this case by hand-tuned exception; v8's ΔW_R-CUSUM catches it with the same detector grammar as the epidemic case — one alarm theory, no special-casing.

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11. Validation Program

11.1 Current Status, Stated Plainly

The model is implementable but not yet an empirically mature standard. Real-world autonomous propagators remain rare relative to historical malware classes; the framework cannot claim validation against a large corpus of confirmed incidents, and this paper declines to imply otherwise. The honest pathway at this stage is synthetic exercises, historical replay, adversarial red-teaming, and instrumented pilots. v8 sharpens what validation must show: the framework now stands or falls on a falsifiable prediction, not on definitional elegance.

11.2 The Core Hypothesis

> **H-LEAD:** For ANP-class activity, P-phase detection (P8) yields measurable lead time before S-phase structural impact — E[L] > 0 where L = t_{S-onset} − t_{forecast} — and this lead time exceeds the warning provided by conventional lateral-movement detection baselines on identical telemetry:
>
> **E[Lead Time_ANP] > E[Lead Time_LateralMovement]**

with the companion claim that containment keyed to forecast determinations reduces S-phase blast radius relative to containment keyed to S-phase detection. Refutation conditions are explicit: if forecast determinations show no positive lead time, or no advantage over baselines, or if predictive coherence fails systematically (impact sets do not overlap exploration sets beyond chance), the predictive layer is refuted and the framework reverts to v7's descriptive taxonomy. A negative result is publishable and the program is designed to produce it cleanly if it is true.

11.3 Metrics

Predictive value (RQ5, primary): mean and distribution of lead time L; blast-radius reduction (contexts impacted under forecast-keyed vs. impact-keyed containment);

false-forecast rate (forecast episodes with no confirmed S-phase within the coherence window, decomposed where determinable into early-containment successes, false alarms, and indeterminate); forecast expiry statistics.

Detection quality (RQ1, RQ3): true-positive rate on controlled exercises; false-positive rate per thousand hosts/workloads; time-to-detect versus Sigma-style correlation baselines on identical telemetry.

Operational burden (RQ4): candidates surfaced per period; median investigation time; downstream response burden.

Coverage and ablation (RQ2): per-primitive availability across tiers; ablation deltas (Section 11.5); statistic-vector component contributions (which of Ḣ_X / ΔW_R / D fired, per detection).

Containment value: time-to-containment improvement; recovery cost under rollback paths; P-band versus S-band action collateral measurements (the Section 8.6 economics, measured).

11.4 The Reference Corpus

An open synthetic-trace corpus mirroring the conformance classes, *phase-labeled* in v8:

- **Positive:** epidemic and surgical propagation across the three telemetry tiers, with ground-truth clause labels and **phase boundaries** (P-onset, S-onset per trace — the labels the lead-time metric requires).
- **Negative:** orchestration storms, patch rollouts, autoscaling bursts, CI/CD fan-out, vulnerability scanning, backup sweeps — with declared-automation inventories and change tickets included, so attribution mechanics are exercised, not assumed.
- **Ambiguous:** slow-and-low (CUSUM-regime), automation mimicry (divergence-regime), surgical concentration (ΔW_R-regime), noise-cloaked spread inside churn, dormancy-gapped P→S campaigns, and out-of-band-primed attacks with minimal P-phase (the stealth bound, measured).

Traces are generated in instrumented containerized topologies (~50–200 simulated contexts) with weighted context inventories, normalized to the event schema, published with generation scripts. **Dual-use posture:** the corpus records behavioral telemetry *shapes* — event sequences, graph structures, timing and weight distributions — not exploit code, payloads, or operational tooling. Publishing what propagation looks like in telemetry arms defenders; attackers do not need a defender's corpus to know what their own tools do.

11.5 Preliminary Synthetic Study (v8.1)

To move the framework from specification toward experimental system, v8.1 reports a deliberately small synthetic study: a minimal implementation of the P8 detector (statistic vector; spike thresholds with heavy-tail quantile substitution; strictly conservative CUSUM; multi-scale pooled windows at 20× and 60×) run against four scenario

generators in a 150-context environment with lognormal weights, five high-value contexts, a declared-automation template, and attribution-leakage baseline noise. Thirty seeded replicates per scenario; ~200 windows per run; calibration on 600 baseline windows per replicate. Code and parameters ship with the paper (anp_sim).

Results.

Scenario	Detected	Forecast before impact	Median lead (windows)	Dominant evidence
Epidemic (branch 3/window)	30/30	29/30	4	ΔW_R (19), D (11)
Surgical (3 high-value targets)	29/30	29/29	5	ΔW_R (27)
Slow-and-low (0.2× rate)	12/30	12/12	26	ΔW_R (10), D (2)
Patch storm, good attribution	—	—	—	5/30 false forecasts
Patch storm, degraded attribution	—	—	—	2/30 false forecasts

Ablation (surgical scenarios): entropy-only detector 0/30; full statistic vector 29/30 — empirical confirmation of the Section 6.3 structural-blindness argument: no parameterization of an entropy threshold detects concentrated high-value acquisition, while ΔW_R catches it in nearly every replicate.

Findings beyond the headline numbers. (1) *The rate-detectability frontier is real and now measured rather than asserted:* at one-fifth the epidemic exploration rate, detection fell to 12/30 — the patient adversary genuinely sinks below attribution-noise levels, surviving by exploring at a rate within baseline drift, exactly the trade-off Section 6.1 prices; detected slow-low runs still yielded a median 26-window lead. (2) *Calibration degeneracy is a first-class failure mode:* early iterations of the study produced 100% false-forecast rates because baseline calibration failed to exercise the volume, reach-gain, and divergence-supported regimes — statistics whose baseline variance collapses to zero become hair triggers; this finding is now a conformance requirement (Sections 4.4, 5.1). (3) *Shape-consistent attribution leakage is benign:* sustained template-shaped leakage during degraded-attribution storms did not elevate false forecasts under quantile calibration — divergence stays low when the leak looks like the template — relocating the attribution-fragility risk specifically to shape-anomalous leakage. (4) *Divergence statistics require support gating:* KL estimates on small residual windows are noise; minimum-support thresholds and Dirichlet smoothing are implementation requirements, not refinements.

Caveats, prominently. This is a toy generator: scenario dynamics are simple stochastic processes, not red-team behavior; the phase-label circularity threat (Sec-

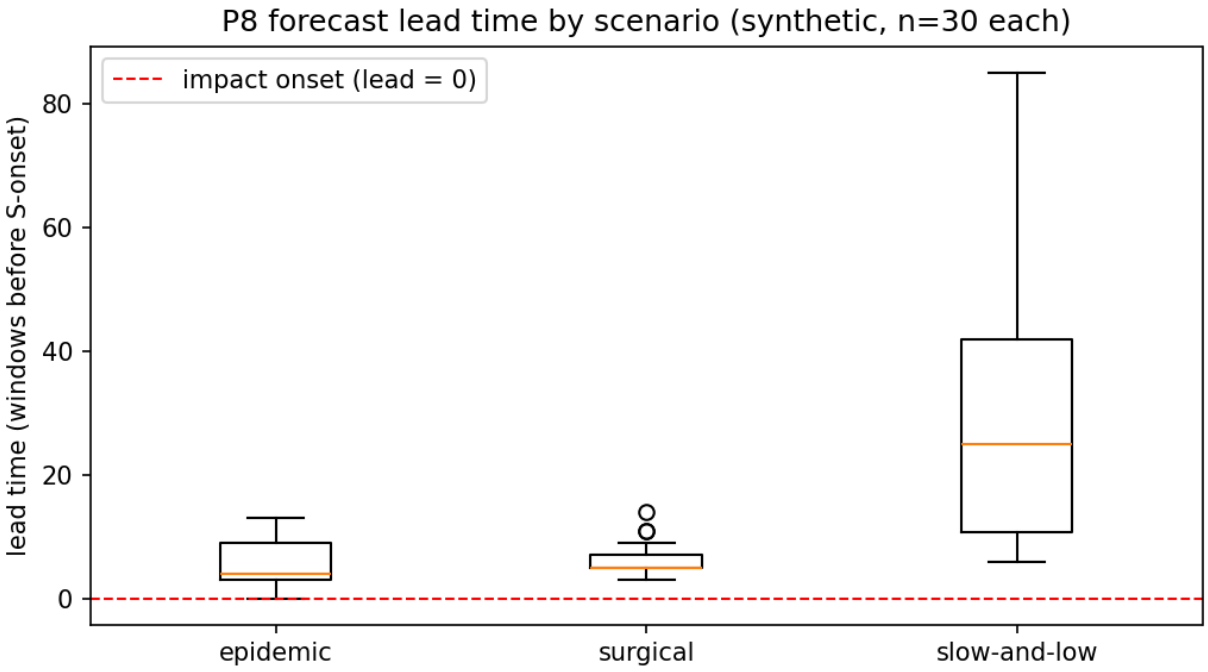


Figure 1: Forecast lead time by scenario

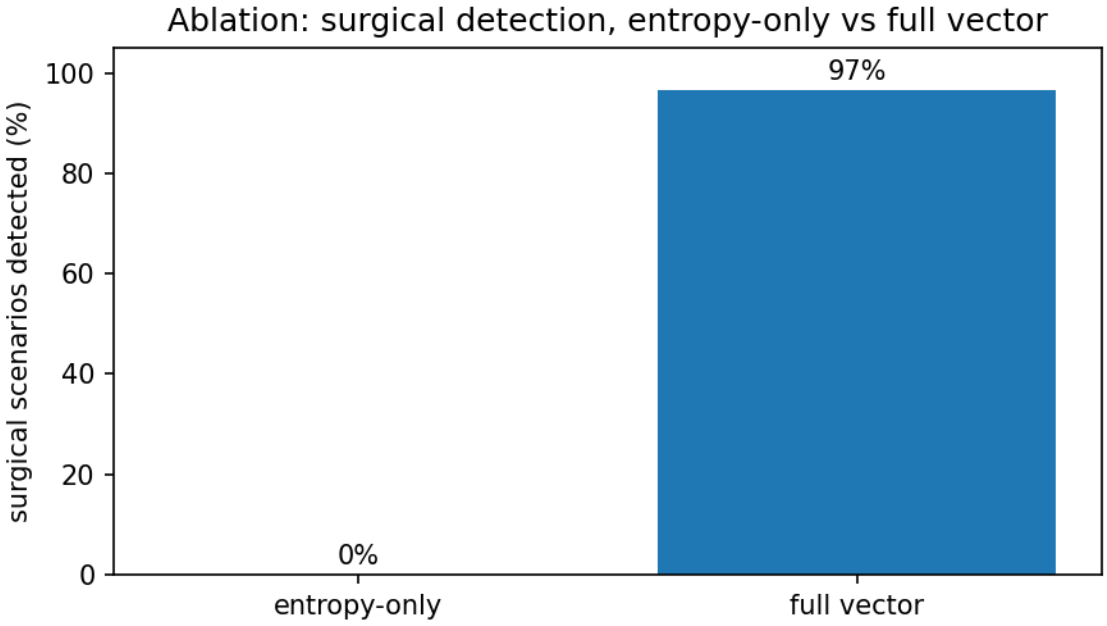


Figure 2: Ablation: surgical detection

tion 11.7) applies in full — the generators implement the P→S structure the detector assumes, so these results demonstrate *pipeline correctness and statistic behavior*, not real-world efficacy; lead times are in abstract windows; and a single environment parameterization was used. The study's purpose is to convert the specification into a running experimental system, expose implementation requirements (it exposed four), and provide the baseline harness the full corpus program (Section 11.4) extends. It succeeds at exactly that scope and claims nothing beyond it.

11.6 Study Designs

Lead-time study (RQ5, primary). On phase-labeled positive and ambiguous traces: run P8 with calibrated parameters; record forecast times; compare against S-onset ground truth and against baseline detector first-alert times. Report lead-time distributions per detection mode (spike/CUSUM/surgical) and per telemetry tier. Containment arm: replay traces under forecast-keyed versus impact-keyed response policies in the simulation topology; measure blast-radius deltas. Out-of-band-primed traces bound the coverage claim honestly.

Primitive ablation (RQ2). Each primitive alone, pairwise, and in band configurations; v8 adds component-level ablation *within* P8 (entropy-only vs. +ΔW_R vs. +D vs. full vector) — the experiment that directly validates Section 6's constructive argument that the vector is necessary.

Reasoning comparison (RQ1, RQ3). Model-assisted cross-plane reasoning — frontier models given normalized primitive outputs and event windows, tasked with clause-by-clause classification and evidence citation — against tuned correlation baselines on identical inputs. Graded on classification accuracy *and* evidence fidelity (do cited events support the cited clause?). Ambiguous-class traces are the adversarial stressor.

Analyst explanation study (RQ4). Practitioners triage surfaced candidates with and without evidence-linked explanations; measure investigation time, decision accuracy, trust calibration. Gates the interpretability commitment against the possibility that explanations merely feel useful.

Pilot deployments. At least two operational settings — one rich-telemetry, one metadata-limited (ANP-N) — instrumented against the 11.3 metrics, with calibration histories preserved as conformance evidence. Pilots measure the quantity simulation cannot: real declared-automation inventories, real baseline drift, real false-forecast economics.

11.7 Threats to Validity

- **Synthetic-real gap:** corpus traces encode our model of the threat; a real propagator may exhibit structure the generators lack. Mitigations: adversarial red-team trace contributions; published generator parameters for community extension; pilots as external check.
- **Phase-label circularity (new, v8-specific):** if corpus generators implement the P→S structure the detector assumes, lead-time results partially measure the generator, not the world. Mitigations: red-team traces generated *without* phase

instructions, phase-labeled post hoc by independent annotators; pilot incidents as the decisive evidence; circularity prominently disclosed in all results.
- **Baseline fairness:** correlation baselines must be tuned with effort comparable to the primitives, or RQ3/RQ5 comparisons overstate the framework. Mitigation: published baseline-tuning protocol.
- **Calibration overfitting:** parameters tuned on the corpus may not transfer; cross-environment calibration transfer is itself a reported experiment.
- **Evaluator leakage:** models evaluated on the corpus may have trained on this paper or artifacts. Mitigations: held-out generation with novel parameterizations; results reported by corpus-release epoch.
- **Weight-assignment subjectivity (new):** ΔW_R inherits the context-weight model; poorly assigned weights manufacture or suppress surgical detections. Mitigations: weight sensitivity analysis as a standard reported experiment; default weight derivation from measurable graph properties (privilege out-degree) rather than manual sensitivity labels alone.

11.8 Roadmap

1. **Publish ANP v1.0 specification** with the seismic layer integrated (this document is the candidate text)
2. **Release the phase-labeled reference corpus** with generation scripts and declared-automation inventories
3. **Build reference implementations** — P8/P9 first (they are the v8 novelty and the cheapest to validate), then the carried library
4. **Run the lead-time study and ablations** (RQ5, RQ2); publish results and the benchmark
5. **Run reasoning comparison and analyst studies** (RQ1, RQ3, RQ4)
6. **Validate operationally** in two pilot environments; **stand up governance** for change proposals, profiles, and conformance claims

11.9 Limitations

- **Empirical scarcity:** real-world incidents are not yet numerous enough for large-scale validation
- **Stealth bound:** fully linear, out-of-band-primed propagation produces no detectable P-wave (Section 6.5) — coverage is in-band exploration, stated and measured
- **Graph dependency:** the seismic layer requires maintaining G(t) and an attribution inventory; environments that cannot, fall back to fixed thresholds with reduced robustness
- **Baseline criticality:** every detector statistic is baseline-relative; poor calibration or unrecorded baseline drift degrades both sensitivity and specificity
- **Calibration complexity:** k, κ, h, δ, weights, and phase windows require environment-specific tuning and ongoing recalibration
- **Adversarial adaptation:** detection logic must evolve continuously against an adversary that reads the specification
- **Privacy trade-offs:** cross-tenant correlation holds collective defense in tension with data sovereignty

- **Attribution boundary:** ANP describes propagation behavior, not responsibility; actor attribution remains a separate discipline

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12. Research Agenda

1. **Empirical incidence tracking.** A public, privacy-preserving registry of candidate ANP incidents — converting empirical scarcity from a limitation into a measured quantity, and providing the earliest signal of class emergence.
2. **Privacy-preserving correlation at scale.** Whether secure aggregation over the (naturally low-dimensional) P-phase statistics detects synchronized weak signals across hundreds of tenants at acceptable cost and leakage.
3. **Adversarial co-evolution.** Structured red-team competitions against the corpus — attackers extend generators, defenders extend primitives — to surface evasion families this paper has not imagined, and to supply the non-circular phase-labeled traces Section 11.6 requires.
4. **Orchestration integration.** Mapping phase-keyed response onto existing SOAR/XDR policy surfaces, so adoption is configuration rather than platform replacement.
5. **Theory of the entropy bound.** Formal characterization of the minimum in-band exploration an adaptive propagator must perform to achieve given objectives under given prior knowledge — the information-theoretic floor under the P-wave, which would convert Section 6.5's trade-off argument from empirical claim to theorem and define the stealth bound exactly.
6. **The economics of machine-speed defense.** Formalizing Section 8.6 with measured C_fp and C_fn from pilots — a principled answer to how much autonomy to grant defensive containment, a question that generalizes beyond this framework.

The fifth and sixth directions state the through-line: ANP is a case study in governing autonomous systems with autonomous systems — bounded, evidenced, reversible, accountable. The propagator and the containment loop are the same species of object pointed in opposite directions, and the disciplines this framework imposes on defense — interpretability, corroboration, rollback, human-authorized escalation, and now prediction with stated refutation conditions — are a working sketch of responsible autonomy under adversarial pressure.

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13. Conclusion

Version 8.0 shifts the Autonomous Network Propagator framework from categorization to prediction. The class is no longer defined only by what its members are — autonomous, self-propagating, boundary-crossing, coherent — but by what they must do: learn before they spend. The seismic model formalizes that obligation as a detectable precursor: exploration-entropy expansion, weighted reach gain, and distributional divergence beyond calibrated baselines, accumulated by CUSUM against patience, checked against human branching envelopes, vetoed by declared-automation

attribution, and bound to subsequent impact by predictive coherence with an explicit null model. Classification splits into forecast and confirmation so that early warning remains early; response is phase-keyed so that containment is cheapest exactly when it is earliest; and the framework's value is denominated in a measurable currency — lead time — with stated refutation conditions.

The six-clause definition, nine-primitive library, response governance, ANP-N variant, conformance and governance mechanisms, and ethical boundaries stand as the structure inside which the predictive layer operates. The stealth bound is stated rather than hidden: a propagator that forgoes in-band exploration evades the P-wave and forfeits the adaptive advantage that defines the class — a trade-off imposed, priced, and accepted.

The framework now makes a single load-bearing promise: detect the learning, and you gain time against the spending. Whether that promise holds is RQ5, and the validation program is built to answer it — including against us. We invite the research and practitioner communities to engage the framework, attack its statistics, extend its corpus, refute its hypothesis if it can be refuted, and govern its evolution.

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Appendix A — Revision Ledger (v7.0 → v8.0)

Carried per ThinkWell Labs revision-ledger practice. All v7.0 structural elements preserved; v8.0 extends, corrects, and re-grounds without structural compression.

#	Change	Type
1	Seismic two-phase model (P-phase/S-phase) introduced as predictive core; lead-time hypothesis (H-LEAD) made the framework’s central empirical claim; RQ5 added	Addition (core)
2	Environment formalized as directed multigraph G(t) with typed edges and attribution sets V_A (declared automation), V_H (ticketed change); residual-statistics-after-attribution made the computation basis	Addition (formal)

#	Change	Type
3	**Correction:** reachable-set entropy conflated potential with activity — split into observed-contact exploration entropy H_X (detection) and weighted reach W_R over $R(v_0,t)$ (response/forecast)	Correction
4	**Correction:** surgical-attack claim that weight concentration produces an entropy spike was directionally wrong (concentration lowers Shannon entropy); surgical detection re-grounded on weighted reach gain ΔW_R	Correction
5	**Correction:** cumulative integral $\int \dot{H}\, dt$ telescopes to net change and is the wrong accumulator; slow-and-low defense re-grounded on CUSUM (Page 1954) with reference value κ and decision interval h	Correction
6	**Correction:** classification rule requiring S-phase observation forfeited early-warning value; determination split into ANP-forecast / ANP-confirmed (four-valued with candidate / not-ANP)	Correction (logic)

#	Change	Type
7	Detection statistic vector defined (Ḣ_X breadth, ΔW_R value capture, D_KL/spectral shape divergence); constructive necessity argument via stress tests; component ablation added to validation	Addition
8	A4 rewritten from fixed velocity thresholds to calibrated stochastic bounds (spike μ+kσ / quantile, CUSUM, surgical); legacy v6/v7 thresholds demoted to fallback calibration profile	Revision (clause)
9	Human branching bound formalized (B(t) > E[B]_hum + k_B·σ_B, peer/time-conditioned, attribution-gated) as behavioral A1 evidence; workforce-monitoring repurposing prohibited (§9.7)	Addition + ethics
10	Predictive coherence formalized with explicit null model (weighted hypergeometric E_rand, margin δ, significance); integrated as temporal dimension of A5; “declared automation is deterministic” softened to template-conservative (stochastic scale, conservative shape)	Addition + correction

#	Change	Type
11	Stealth limitation broadened: out-of-band knowledge acquisition (IaC exfiltration, purchased maps) bypasses P-phase, not only “perfect stealth”; coverage bound stated and measured in corpus	Honesty extension
12	Two primitives added: P8 entropy gradient (P-phase detector, A4/A1), P9 predictive coherence (confirmation only); availability matrix extended; P7/ΔW_R potential-vs-acquired distinction made precise	Addition
13	Minimum telemetry reduced to identity transitions, network adjacency, control-plane modifications, change-ticket metadata; ANP-N strengthened with breadth-only predictive capability	Revision (requirements)
14	Response mapping phase-keyed (P-band reversible / S-band full ladder); forecast expiry with auto-rollback added; containment economics extended to phase split	Addition
15	Attribution inventory added to poisoning threat model; honey-resource synergy with P-wave SNR noted under deception guardrails	Extension

#	Change	Type
16	Vignettes re-evaluated under v8 statistics with phase annotations, measured lead time (A), residual-attribution mechanics (B), ΔW_R-CUSUM unification of the surgical case (C)	Revision
17	Validation program rebuilt around H-LEAD: lead-time study primary; false-forecast decomposition; phase-label circularity and weight-subjectivity added to threats to validity; phase-labeled corpus	Revision (program)
18	Early-warning analogies sectioned (seismology, CUSUM/SPC, epidemic surveillance) with analogy bounds stated; references added: Allen & Melgar, Page, Shannon, Kullback–Leibler	Addition
19	Log base fixed (bits, $\log_2$); notation collision resolved (V_H for ticketed contexts vs. H for entropy); E_rand defined; all "□" conditions formalized as μ+kσ bounds	Hygiene

No v7.0 clause, primitive, response element, exclusion, governance mechanism, ethical boundary, or limitation was removed or weakened. Three v8-source mathematical claims were corrected before integration (ledger items 3–6); the corrections strengthen the framework the claims were intended to support.

Appendix B — Revision Ledger (v8.0 → v8.1)

#	Change	Type
1	Proposition 1 (exploration lower bound) added via Fano's inequality, with Corollaries 1 (stealth priced in priors) and 2 (adaptivity premium); scope of what it does/does not establish stated (§5.10); §6.5 grounded on it	Addition (theory)
2	Principled default weight derivation added: w_i from privilege out-degree × control-plane authority × sensitivity; adversary-awareness argument; weight sensitivity analysis made a required reported experiment (§5.2)	Addition (normative)
3	Attribution maturity stated as operational precondition with specified degraded-mode behavior (a–d); minimum attribution bar for forecast-grade operation (§5.1)	Honesty extension
4	Box 1 comparison table: ANP v8 vs lateral-movement detection (§3.8)	Addition (exposition)
5	Preliminary synthetic study added (§11.5): n=30/scenario, lead-time distributions, full-vector vs entropy-only ablation (29/30 vs 0/30 surgical), false-forecast rates, rate-detectability frontier; toy-generator caveats and circularity disclosure prominent	Addition (empirical)

#	Change	Type
6	Four implementation requirements promoted from study findings: calibration must exercise all detector regimes (degeneracy = hair trigger); divergence requires support gating + smoothing; multi-scale pooled windows for slow shifts; strictly conservative CUSUM (κ at baseline max) for bounded false-alarm drift (§4.4, §5.1, §11.5)	Normative (from experiment)
7	Empirical nuance recorded: shape-consistent attribution leakage does not elevate false forecasts; fragility risk relocated to shape-anomalous leakage (§5.1, §11.5)	Correction (narrative)
8	Section renumbering: former §11.5–11.8 → §11.6–11.9; reference [21] (Cover & Thomas) added	Hygiene

End of document.